

Generalized method of a resolvent operator expansion

M. Znojil

Institute of Nuclear Physics, Czechoslovakian Academy of Sciences, Řež, Czechoslovakia
(Received 9 October 1975)

The method of inversion of a quasitridiagonal matrix is suggested. The more general interpretation of the resulting expansion of the resolvent operator on a vector is given that does not use the concept of matrix elements nor inner product in a vector space. From this point of view, the linear operator $\hat{H}$ is employed recursively to define the basis starting with M initial vectors. The expansion of the action of the resolvent operator $\hat{R}(E) = (E - \hat{H})^{-1}$ on any basis vector is given in the same basis with simply calculated coefficients, thus generalizing the usual power series expansion.

1. INTRODUCTION

Matrices $\bar{H}$ that in some basis for some $p > 0$ satisfy the condition

$$\langle n | \bar{H} | n + l + p \rangle = \langle n + l + p | \bar{H} | n \rangle = 0, \quad l = 1, 2, \dots, \quad (1)$$

are often useful in physics (e.g., kinetic energy operator or local potential operator of the form $\bar{U}(r) = \sum_{k=0}^p a_k r^{2k}$ in the harmonic oscillator basis.¹) At the same time, $\bar{H}$ can appear in the form of the resolvent operator (e.g., propagator) $\bar{R}(E) = (E - \bar{H})^{-1}$. In the first part of the present paper (Sec. 2) we use a slight generalization of the condition (1) and suggest the method of calculation of the matrix elements of $\bar{R}(E)$.

In the second part of this paper (Sec. 3), we give another more general interpretation of the formulas. We take any linear operator $\hat{H}$ in a vector space V and define the basis in V by the repeated action of $\hat{H}$, starting from p initial vectors $|X_1\rangle, \dots, |X_p\rangle$. In this basis we then expand the action of the resolvent operator $\hat{R}(E) = (E - \hat{H})^{-1}$ on any basis vector. The basis need not be orthonormal, but it will be complete in a subspace $\bar{V} \subset V$ invariant with respect to $\hat{H}$, $\hat{H}\bar{V} \subset \bar{V}$ (cf. Ref. 2, which we shall denote as I).

The results may be looked upon as a generalization of I, and therefore also as a generalization of the expansion of Haydock³ or that of the power series expansion of $\hat{R}(E)$.

2. QUASITRIDIAGONAL MATRIX INVERSION

We call the matrix H tridiagonal, when it fulfills the condition (1) with $p = 1$,

$$H_{k, k+l+1} = H_{k+l+1, k} = 0, \quad k, l = 1, 2, \dots$$

Let us now have a N -dimensional tridiagonal matrix H , the elements of which are not numbers, but M -dimensional matrices $H_{k, l}$, $k, l = 1, 2, \dots, N$. We shall call the $(M \times N = N_D)$ -dimensional matrix H quasitridiagonal. In the matrix notation, we have

$$H_{k, k+l+1} = H_{k+l+1, k} = 0, \quad l = 1, 2, \dots, N - k - 1, \quad k = 1, 2, \dots, N - 2, \quad (2)$$

and denote the transposed nonzero matrices as follows

$$\begin{aligned} A_k &= H_{k, k}^T, \quad k = 1, 2, \dots, N, \\ B_k &= H_{k+1, k}^T, \\ C_{k+1} &= H_{k, k+1}^T, \quad k = 1, 2, \dots, N - 1. \end{aligned} \quad (3)$$

We shall denote the M -dimensional vectors and matrices by the boldface and sans serif fonts, respectively, and

write the indices of their elements on top; below we denote the different M -dimensional matrices while the number in the matrix H is identified by the pair of indices. We define the basis in the form of the column vectors X_k^r ,

$$(X_k^r)_i = \delta_{i, (k-1)M+r}, \quad i = 1, 2, \dots, N_D, \quad r = 1, 2, \dots, M, \quad k = 1, 2, \dots, N,$$

in the N_D -dimensional vector space V_{N_D} . The matrix H acts on the vector X_k^r in the space V_{N_D} according to the prescription

$$HX_k^r = \sum_{s=1}^M [C_k^{rs} X_{k-1}^s + A_k^{rs} X_k^s + B_k^{rs} X_{k+1}^s]. \quad (4)$$

We shall write (4) also in the abbreviated matrix notation as

$$HX_k = C_k X_{k-1} + A_k X_k + B_k X_{k+1}.$$

Now, we intend to find the resolvent matrix $R(E) = (E - H)^{-1}$, where E is a variable not equal to any eigenvalue of H . Matrix elements of $R(E)$ can be defined in our basis as the projections of the vectors

$$Y_k^r = R(E) X_k^r. \quad (5)$$

If we use the expansion

$$Y_k^r = \sum_{l=1}^N \sum_{s=1}^M D_{kl}^{rs} X_l^s = \left(\sum_{l=1}^N D_{kl} X_l \right)^r, \quad (6)$$

then we have

$$\langle X_k^r | R(E) | X_l^s \rangle = D_{kl}^{rs} \quad (7)$$

due to the orthogonality of the basis vectors. Since

$$\begin{aligned} X_k^r &= (E - H) Y_k^r = E \sum_{ls} D_{kl}^{rs} X_l^s - \sum_{ls} D_{kl}^{rs} H X_l^s \\ &= \sum_{lst} [D_{kl}^{rt} (\delta_{st} E - A_l^{ts}) - D_{kl+1}^{rt} C_{l+1}^{ts} - D_{kl-1}^{rt} B_{l-1}^{ts}] X_l^s, \end{aligned} \quad (8)$$

i.e.,

$$X_k = \sum_l [D_{kl}(E - A_l) - D_{kl+1} C_{l+1} - D_{kl-1} B_{l-1}] X_l,$$

we get in view of the independence of X_k^r the relation for D 's,

$$\delta_{rs} \delta_{kl} = \sum_t [D_{kl}^{rt} (\delta_{st} E - A_l^{ts}) - D_{kl+1}^{rt} C_{l+1}^{ts} - D_{kl-1}^{rt} B_{l-1}^{ts}], \quad (9)$$

where δ_{kl} is a Kronecker delta and $D_{k0}^{rs} = 0$ is added for convenience. Since we have (4) and (6), we may expand

$$\begin{aligned} X_k^r &= \frac{1}{E - H} E X_k^r - \frac{1}{E - H} H X_k^r \\ &= E Y_k^r - \sum_s [C_k^{rs} Y_{k-1}^s + A_k^{rs} Y_k^s + B_k^{rs} Y_{k+1}^s], \end{aligned} \quad (10)$$

i.e.,

$$X_k = \sum_i [E D_{ki} - A_k D_{ki} - B_k D_{k+1,i} - C_k D_{k-1,i}] X_i$$

and get the second relation for D 's,

$$\delta_{rs} \delta_{kl} = \sum_i [(\delta_{si} E - A_k^{rt}) D_{ki}^{ts} - B_k^{rt} D_{k+1,i}^{ts} - C_k^{rt} D_{k-1,i}^{ts}], \quad (11)$$

which may be used in the recurrent determination of the values D_{ki}^{rs} . We put

$$D_{k+1,i}^{rs} = \sum_t D_{ki}^{rt} \alpha_{i+1}^{ts}(k), \quad (12)$$

$$\alpha_i^{ts}(k) = \sum_{r=1}^M B_{i-1}^{rt} F_i^{rs}, \quad l \geq k > 0, \quad B_0^{ts} = \delta_{ts},$$

in complete analogy with I. The method of proving is that used in the paper I (mathematical induction), the only change being the use of the noncommuting matrices instead of the bare numbers. The insertion of (12) into (9) leads to the recurrent relation

$$\sum_s F_k^{rs} [\delta_{st} E - A_k^{st} - \sum_v \alpha_{k+1}^{sv} C_{k+1}^{vt}] = \delta_{rt} \quad (13)$$

i.e.,

$$F_k = [E I - A_k - \alpha_{k+1} C_{k+1}]^{-1}$$

together with the boundary condition

$$\alpha_{N+1} = 0 \quad (14)$$

which follows from the finite dimension of the matrix H . The evaluation of the sequence F_l , $l = 1, 2, \dots, N$ is, according to (13), the matter of the M -dimensional matrix inversion, which is N times repeated. It follows from the boundary condition (14) that $\alpha_i^{rs}(k)$ does not depend on the index in the parentheses, $\alpha_i^{rs}(k) = \alpha_i^{rs}$, so that the sequence F_l is universal for all the vectors Y_k and is used repeatedly in formula (12). This is the main advantage of the calculation method and the reason for saving a number of operations.

The second advantage of the method manifests itself in approximate numerical calculations when the number N is only the cutoff parameter corresponding to the truncation of an infinite basis. Then the limit $N \rightarrow \infty$ should be taken in principle, while the number of the calculated $R(E)$ matrix elements may remain fixed. In practice, we always stop at some finite N . In our case, the amount of necessary work is proportional to N , while in the inversion of the general N_D -dimensional matrix $(E - H)$ the number of operations is proportional to N^3 (see Ref. 4). We are able to work with a higher N and therefore reach higher precision. At the end of this section, we add some other useful formulas. Since the ansatz

$$D_{k+1,i}^{rs} = \sum_l \gamma_{k+1}^{rl}(l) D_{ki}^{ls} \quad (15)$$

leads to the relation

$$\gamma_{k+1}^{rt}(l) = \gamma_{k+1}^{rt} = \sum_s F_k^{rs} C_{k+1}^{st} \quad (16)$$

according to (11), we save further operations and conclude that only the calculation of the D_{kk}^{rs} matrices necessitates the use of the recurrence relations (9) or (11). The insertion of the relations (12) and (15) into (9) provides the relation

$$D_{kk}^{rs} = \sum_t [\delta_{rt} + \sum_{uv} F_k^{ru} C_k^{uv} D_{k-1,k-1}^{vw} B_{k-1}^{wt}] F_k^{ts} \quad (17)$$

i.e.,

$$D_{kk} = [I + F_k C_k D_{k-1,k-1} B_{k-1}] F_k.$$

Thus, the full solution of the problem of finding the matrices D_{ki}^{rs} in the expansion (6) is represented by the recurrence relations (13) together with the initialization (14) and the relations (17) and (12) and (15), (16).

3. THE CASE OF A GENERAL OPERATOR

Let us suppose that we have a linear operator $\hat{H}$ in a vector space V and we choose the set of M initial vectors $|X_1^r\rangle \in V$, $r = 1, 2, \dots, M$. We then produce the sequence of the vectors $|X_k^r\rangle$, $r = 1, 2, \dots, M$, $k = 2, 3, \dots$ according to the definition

$$|X_{k+1}^r\rangle = \sum_s (B_k^{-1})^{rs} \left\{ \hat{H} |X_k^s\rangle - \sum_t [C_k^{st} |X_{k-1}^t\rangle + A_k^{st} |X_k^t\rangle] \right\}, \quad (18)$$

where the sequences of numbers

$$A_k^{rs}, B_k^{rs}, C_{k+1}^{rs}, \quad r, s = 1, 2, \dots, M, \quad k = 1, 2, \dots, \quad (19)$$

are chosen arbitrarily provided that the $M \times M$ matrices B_k can be inverted. The choice of the sequences (19) for fixed number M should make the structure of the new vectors (18) sufficiently simple for practical purposes.

The formalism of the previous section is applicable. The definition (18) is equivalent to the relation (4). For the operator $\hat{R}(E) = (E - \hat{H})^{-1}$ we may define the vectors (5), and the expansion (6) can then be written

$$|Y_k^r\rangle = \hat{R}(E) |X_k^r\rangle = \sum_{is} D_{ki}^{rs} |X_i^s\rangle. \quad (20)$$

The new basis vectors need not be orthonormalized so that the relation (7) need not be valid, but the derivation of (9) and (11) may proceed in the same way. Therefore, the relations (12), (15), (16), and (17), together with (13) and (14), may be used for the calculation of the expansion coefficients D_{ki}^{rs} at some cutoff N . When the value of some $D_{k_0 l_0}^{r_0 s_0}$ becomes independent of the cutoff N , the exact result is obtained provided that the expansion (20) is convergent.

The general proof of the convergence of (20) is especially simple in the $M = 1$ case. It is thoroughly discussed in I. For $M > 1$, complications arise only in view of the matrix character of the expansion. E.g., orthogonality of $|X_k^r\rangle$, at least for different lower indices, can sometimes be introduced and exploited in a way analogous to I. The general formal discussion of the convergence properties is of less value here because of the great number of free parameters. It should be performed for every special case separately, along the same general lines as in I.

4. SUMMARY

The main result of this paper is the expansion (20) expressing the action of a resolvent operator $\hat{R}(E) = (E - \hat{H})^{-1}$ on a vector. This expansion is characterized by a set of free input parameters with twofold significance. First, they enable us to define (and simplify) the basis vectors in a subspace of the given vector space invariant with respect to the action of the operator $\hat{H}$. Second, they appear in calculation of the expansion coefficients.

There exist two extreme possibilities:

(1) We choose trivial input parameters $M=1$, $B=1$, $A=C=0$. Then the expansion reduces to the usual power series expansion of the resolvent operator, i.e., the calculation of the expansion coefficients becomes trivial. The expansion vectors are not simplified at all.

(2) We choose some most elementary expansion vectors in advance. The input parameters must then be put equal to the matrix elements of the operator $\hat{H}$ the way it was done in Sec. 2. If there are no zero matrix elements, the formulas may be used only with $N=2$. The calculation of the expansion coefficients is very involved in this case, being practically equivalent to the whole matrix inversion.

Therefore, the presented formalism constitutes an appropriate link between these two extremes. It is very flexible and enables us to find an optimal compromise between the simplicity of the expansion basis vectors and that of the expansion coefficients calculation. In

practice, it seems to be most adequate for treating the operator $\hat{H}$ with the largest matrix elements lying near the main diagonal in some reasonable basis.

Note added in proof: The range of applicability will be enlarged when the dimensions $M = M_k$ of submatrices in the definition (3) are not equal to the same number and depend on the index k . The multiplication of (not always square) submatrices remains well defined for $M_1 \neq M_2 \neq \dots$, and the whole formalism is valid even in this case. Further, the method may be adapted for the operator diagonalization (paper submitted to Can. J. Phys.).

¹S. G. Nilsson, Kgl. Danske Vidensk Selsk., Mat.-Fys. Medd. **29**, 1 (1955).

²M. Znojil, J. Phys. A: Math. and Gen. **9**, 1 (1976).

³R. Haydock, J. Phys. A: Math., Nucl. and Gen. **7**, 2120 (1974).

⁴F. S. Acton, *Numerical Methods That Work* (Harper I. E., New York, 1970).